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Intensive Blood Pressure Management After Endovascular Thrombectomy: The Impact of White Matter Hyperintensity Severity

Supreme Neurologist Board Level — OPTIMAL-BP Trial Secondary Analysis

Specialty: Neurology — Cerebrovascular Disease / Critical Care Neurology
Topic: Intensive vs Conventional Blood Pressure Targets Following Successful Endovascular Thrombectomy in Acute Ischemic Stroke — The Modifying Role of White Matter Hyperintensity Severity
Date: March 2026
Evidence Level: Secondary Analysis of Multicenter RCT (OPTIMAL-BP Trial, n=271)
Target Audience: Neurology Residents, Stroke Fellows, Vascular Neurologists, Neurointerventionalists, Neurointensivists, Clinical Researchers, and Board Examination Candidates

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1. Executive Summary

The post-thrombectomy period represents a critical window for blood pressure (BP) optimization, with competing priorities: maintaining adequate cerebral perfusion pressure in the newly reperfused territory while avoiding hyperperfusion injury, hemorrhage transformation, and sustained elevation that may worsen edema. The **OPTIMAL-BP trial** challenged the assumption that intensive BP lowering is universally beneficial following successful endovascular thrombectomy (EVT), demonstrating that targeting systolic BP <140 mmHg within 24 hours worsened functional outcomes compared to conventional management (140–180 mmHg).

A landmark **secondary analysis** of the OPTIMAL-BP trial specifically examined how pre-existing white matter hyperintensity (WMH) severity modifies the relationship between post-EVT BP management and clinical outcomes. In **271 patients** stratified by WMH burden (Fazekas score ≥3 defining severe WMH), the study revealed a critical heterogeneity of treatment effect:

WMH Severity	Intensive BP (<140 mmHg)	Conventional BP (140–180 mmHg)	Outcome
Severe WMH (Fazekas ≥3)	29.9% mRS 0–2	51.3% mRS 0–2	**Intensive BP significantly worse**
Mild WMH (Fazekas <3)	No significant difference	—	**NS between groups**

Key Conclusions: 1. **Severe WMH identifies a vulnerable subgroup** in which intensive BP lowering is harmful 2. **"One size does not fit all"** — uniform intensive BP targets post-EVT are not appropriate 3. **Cerebral autoregulation impairment** in severe WMH explains the observed harm 4. **Individualized BP targets** based on baseline brain health are warranted 5. **Clinical implication:** Avoid intensive BP lowering (<140 mmHg) in patients with severe WMH post-EVT

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2. Background: Stroke, Endovascular Thrombectomy, and Blood Pressure Management

2.1 Acute Ischemic Stroke and Large Vessel Occlusion

Acute ischemic stroke (AIS) due to large vessel occlusion (LVO) represents the most severe stroke phenotype, accounting for approximately 30–40% of all ischemic strokes but disproportionately contributing to mortality, disability, and healthcare burden. The paradigm shift in LVO management occurred with the publication of multiple landmark EVT trials (MR CLEAN, EXTEND-IA, SWIFT PRIME, REVASCAT, ESCAPE) demonstrating that rapid mechanical thrombectomy achieving successful reperfusion (TICI 2b–3) significantly improves functional outcomes when performed within 6–24 hours of symptom onset.

Epidemiology of LVO: - Incidence: ~30–40 per 100,000 population annually - Mortality without treatment: 20–30% at 90 days - With EVT: Mortality reduced to 10–15%; functional independence (mRS 0–2) achieved in 46–51%

2.2 Endovascular Thrombectomy: Technical Considerations

EVT Modalities: | Technique | Mechanism | Reperfusion Rate | |———-|———|————————| | Stent retrievers (Solitaire, Trevo) | Mechanical clot extraction | TICI 2b–3: 59–71% | | aspiration catheter (ADAPT) | Direct aspiration first | TICI 2b–3: 52–82% | | Combined (Solumbra) | Stentriever + aspiration | TICI 2b–3: 70–88% |

Time is Brain: - Estimated 1.9 million neurons lost per minute in LVO - Door-to-groin time: median 60–90 minutes in experienced centers - Successful reperfusion: every 30-minute delay reduces mRS 0–2 by 6–8%

2.3 Blood Pressure Management Post-EVT: The Clinical Challenge

The immediate post-EVT period presents a complex hemodynamic challenge with competing priorities:

Pathophysiologic Considerations:

Factor	Effect	BP Management Implication
Ischemic core	Neuronal death, release of DAMPs	Need adequate perfusion pressure
Penumbra	Hypoperfused but viable tissue	Risk of extension if BP too low
Reperffusion injury	Oxidative stress, inflammation	Suggests BP moderation
Blood-brain barrier disruption	Edema, hemorrhage risk	Suggests BP control
Chronic small vessel disease	Impaired autoregulation	Variable response to BP changes
Collateral circulation	Variable compensatory flow	Patient-specific BP needs

The Autoregulation Curve:

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Cerebral Blood Flow (CBF)
    ↑
    |                    Autoregulation
    |                ┌──────────────────┐
    |                │                  │
    |   ─────────────┤                  ├──────────
    | Normal MAP     │                  |   Critical
    | range          │                  |   Lowering
    | (60-150 mmHg) |                  |
    |                └──────────────────┘
    └──────────────────────────────────────────────→ MAP
        ↓CBF        Optimal Zone        ↑CBF/Edema
       (Ischemia)                      (Hyperemia)
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Traditional BP Targets Post-EVT:

Guideline	Systolic BP Target	Rationale
AHA/ASA 2019	<180/105 mmHg	Expert consensus; limited RCT data
ESO 2021	<180 mmHg	Prevents hemorrhage; maintains perfusion
Current practice	Variable (140–180 mmHg)	Wide variation exists

2.4 The OPTIMAL-BP Trial: Challenging Uniform Intensive Targets

Trial Design: - Multicenter, randomized, open-label, blinded endpoint assessment (PROBE) - 19 stroke centers in South Korea - June 2020 – November 2022 - n=307 patients with LVO, successful EVT (TICI 2b–3), SBP ≥140 mmHg within 2 hours

Randomization: | Arm | Target Systolic BP | Duration | |——|————————-|———-| | **Intensive** | <140 mmHg | First 24 hours | | **Conventional** | 140–180 mmHg | First 24 hours |

Results (Primary Analysis — STOPPED EARLY):

Outcome	Intensive (n=154)	Conventional (n=153)	P-value
mRS 0–2 at 90 days	39.4%	54.4%	P = .02
mRS shift (ordinal)	Worse with intensive	Better with conventional	OR 1.64
Symptomatic ICH	9.1%	4.0%	P = .08
Mortality	17.5%	8.5%	P = .03

Trial stopped early by DSMB for safety concerns.

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3. White Matter Hyperintensity: Pathophysiology and Clinical Significance

3.1 Definition and Imaging Characteristics

White matter hyperintensities (WMHs) are areas of increased signal intensity on T2-weighted and FLAIR MRI sequences within the cerebral white matter. They represent histopathologically:

- **Demyelination** and axonal loss
- **Gliosis** (reactive astrocytosis)
- **Vacuolization** and rarefaction of white matter
- **Small vessel wall thickening** and lipohyalinosis

Clinical significance: - Neuroimaging marker of **cerebral small vessel disease (CSVD)** - Associated with: cognitive decline, gait disturbance, depression, stroke risk - Predicts: post-stroke outcomes, recurrence, mortality

3.2 The Fazekas Scale: Grading WMH Severity

The Fazekas scale is the most widely validated and used visual rating system for WMH burden:

Periventricular WMH (PVH) Scoring:

Grade	Description
0	No periventricular lesions
1	“Caps” or pencil-thin lining
2	Smooth “halo” or irregular periventricular signal
3	Irregular periventricular signal extending into deep white matter

Deep White Matter WMH (DWMH) Scoring:

Grade	Description
0	No deep white matter lesions
1	Punctate foci (≤5 mm)
2	Beginning confluence (6–10 mm, 3+ lesions)
3	Large confluent lesions (>10 mm, extensive)

Total Fazekas Score: PVH + DWMH = 0–6
Severe WMH Definition: Total Fazekas score ≥3

Population Prevalence: - Mild WMH: ~50% of stroke-free elderly - Severe WMH (Fazekas ≥3): ~10–20% of stroke-free elderly; up to 60% in acute stroke populations

3.3 Cerebral Small Vessel Disease: The Underlying Pathology

CSVD refers to pathological processes affecting the small vessels of the brain:

Vessel Type	Size	Pathology	Imaging Findings
Lipohyalinotic arterioles	100–400 μm	Wall thickening, fibrinoid necrosis	WMH, lacunes
Amyloid angiopathy	50–200 μm	Aβ deposition, wall weakening	WMH, microbleeds, lobar ICH
Atherosclerotic	400–800 μm	Plaque, stenosis	WMH, territorial infarcts

Mechanisms of WMH Formation:

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Chronic Hypertension / Aging
          ↓
  Small Vessel Wall Thickening
  (Lipohyalinosis, Fibrinoid Necrosis)
          ↓
  Reduced Cerebral Blood Flow
          ↓
  Ischemia of White Matter
  (Deep Periventricular Territory)
          ↓
  Oligodendrocyte Death
  Myelin Degradation
          ↓
  WHITE MATTER HYPERINTENSITIES
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3.4 Cerebral Autoregulation in CSVD

Cerebral autoregulation maintains constant CBF across a wide MAP range (typically 60–150 mmHg). This mechanism is mediated by:

- **Myogenic response:** Vascular smooth muscle constriction/dilation in response to pressure changes
- **Metabolic factors:** Local metabolites (adenosine, K+, H+) modulate vessel tone
- **Neurogenic factors:** Sympathetic innervation

Autoregulation Impairment in CSVD:

Normal	CSVD State
Wide autoregulatory plateau	Right-shifted, narrowed plateau
Stable CBF 60–150 mmHg MAP	CBF varies with BP changes
Protected perfusion	Vulnerable to hypoperfusion

Right-Shifted Autoregulation:

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CBF
 ↑
 │    Normal          CSVD (Impaired)
 │   ┌───────┐       ┌─────────┐
 │   │       │       │         │
 │───│       │──→────┘         └──────
 │   │       │       ↑           ↑
 │   └───────┘    Higher        CBF
 │  60-150   MAP needed     varies
 └──────────────────────────────────→ MAP
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Clinical Implication: In patients with severe WMH/CSVD: - Higher MAP required to maintain adequate CBF - Aggressive BP lowering → hypoperfusion of vulnerable white matter - Even moderate hypotension can cause ischemia in “oligemic” areas

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4. OPTIMAL-BP WMH Secondary Analysis: Detailed Results

4.1 Study Design and Population

Design: Pre-specified secondary analysis of the OPTIMAL-BP trial
Population: n=271 patients with available baseline MRI and WMH assessment

Baseline Characteristics:

Characteristic	Severe WMH (n=163)	Mild WMH (n=108)
Age, mean ± SD	68.2 ± 10.1	65.4 ± 11.3
Male, n (%)	93 (57.1%)	58 (53.7%)
Hypertension	127 (77.9%)	74 (68.5%)
Diabetes mellitus	54 (33.1%)	29 (26.9%)
Atrial fibrillation	52 (31.9%)	41 (38.0%)
Initial NIHSS, median (IQR)	13 (7–18)	10 (5–16)
ASPECTS, median (IQR)	8 (7–10)	9 (8–10)
IV tPA received	89 (54.6%)	56 (51.9%)
Successful reperfusion (TICI 2b–3)	100%	100%
Time to reperfusion, min	43 (28–72)	38 (25–65)

4.2 Treatment Effect Modification by WMH Severity

Primary Outcome: Functional Independence (mRS 0–2) at 90 Days

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║       PRIMARY OUTCOME: Functional Independence by WMH Severity
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╒
╒  SEVERE WMH (Fazekas ≥3):
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╒  ┌──────────────────────────────────────────────────────────────┐
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╒  │  Intensive BP:     ██████████▒▒▒▒▒▒▒▒▒▒  29.9% (24/80)   │         ╒
╒  │  Conventional BP:  ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒  51.3% (42/82)   │         ╒
╒  │                                                              │
╒
╒  │  Difference:       -21.4 percentage points                   │
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╒  │  P-value:         P = .003                                   │
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╒  │  95% CI:          -35.0% to -7.8%                            │
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╒  MILD WMH (Fazekas <3):
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╒  │  Intensive BP:     ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒  62.3%       │         ╒
╒  │  Conventional BP:  ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒  68.5%     │         ╒
╒  │                                                              │
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╒  │  Difference:       -6.2 percentage points                    │
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╒  │  P-value:         P = .42 (NOT SIGNIFICANT)                  │
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╒  │  95% CI:          -21.2% to 8.8%                            │         ╒
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╒  INTERACTION P-VALUE: P = .008 (significant effect modification)
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Statistical Test for Interaction: - WMH severity × BP treatment group interaction: P = .008 - This confirms that WMH severity **significantly modifies** the treatment effect - The difference is not due to chance

4.3 Secondary Outcomes

Outcome	Severe WMH	Mild WMH
mRS shift (ordinal)	Intensive worse (aOR 2.1, 95% CI 1.2–3.8)	No significant difference
90-day mortality	22.5% vs 11.0% (intensive vs conventional)	No significant difference
Symptomatic ICH	12.5% vs 4.9% (intensive vs conventional)	No significant difference
Excellent outcome (mRS 0–1)	15.4% vs 29.6% (intensive vs conventional)	No significant difference

4.4 Subgroup Analysis Within Severe WMH

Fazekas Score 3 vs 4–6: - Similar direction of effect - Trend toward greater harm with higher Fazekas scores - Limited power for formal statistical testing

Baseline NIHSS Stratification: - Effect consistent across NIHSS severity levels - Trend: greater absolute harm in severe strokes (high NIHSS)

4.5 Mechanistic Interpretation

Why does intensive BP lowering harm patients with severe WMH?

Mechanism	Explanation
Right-shifted autoregulation	Higher BP required to maintain CBF in damaged white matter
Oligemic penumbra extension	Aggressive BP lowering reduces perfusion in watershed zones
Impaired collateral response	Severely affected white matter has reduced collateral capacity
Chronic hypoperfusion	WMH tissue adapted to higher baseline MAP
Blood-brain barrier vulnerability	CSVD causes fragile microvasculature prone to underperfusion

Pathophysiologic Model:

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SEVERE WMH + INTENSIVE BP LOWERING
          ↓
Impaired Autoregulation
(No compensatory vasodilation)
          ↓
Reduced Cerebral Blood Flow in White Matter
(Vulnerable to BP changes)
          ↓
Expansion of Ischemic Injury
(Microinfarcts, Oligemia)
          ↓
Worse Functional Outcomes
(Higher mRS, More Disability)
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5. Clinical Implications

5.1 Implications for Post-EVT BP Management

Revised Algorithm for BP Targets Post-EVT:

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POST-EVT BLOOD PRESSURE MANAGEMENT

Successful Reperfusion (TICI 2b-3)
          ↓
    Check Baseline MRI
    (WMH Assessment)
          ↓
    ┌─────┴─────┐
    │           │
 Fazekas    Fazekas
   <3         ≥3
    │           │
    ↓           ↓
Can consider  AVOID Intensive BP
intensive BP (<140 mmHg)
if clinically ↓
indicated    Target: 140-180 mmHg
          (or higher if tolerated)
```

5.2 Practical BP Targets

Patient Group	Systolic BP Target	Rationale
Severe WMH (Fazekas ≥3)	**140–180 mmHg**	Avoid harm from hypoperfusion
Mild WMH (Fazekas <3)	Individualized; may consider <140 mmHg if tolerated	No significant harm demonstrated
No MRI available	140–180 mmHg	Cautious approach pending evidence
Hemorrhagic transformation	<180 mmHg	Balance perfusion vs bleeding risk
Malignant edema	Individualize; higher targets may be needed	Prevent herniation

5.3 Implications for Stroke Systems of Care

Prehospital/Emergency Department: - Early notification for EVT candidates - Expedited transfer to comprehensive stroke center - BP management en route (avoid precipitous lowering)

Neurointerventional Suite: - Periprocedural BP: Maintain MAP to ensure collateral flow - Post-procedure: Immediate BP assessment and target setting - Consider WMH burden in BP decision-making when MRI available

Neurocritical Care/Stroke Unit: - Continuous BP monitoring (arterial line) - Individualized targets based on WMH severity - Avoid BP variability (U-shaped curve worse outcomes) - Monitor for signs of hypoperfusion (new deficits, imaging changes)

5.4 Imaging Considerations

When is WMH Assessment Available?

Setting	WMH Assessment	Feasibility
Comprehensive stroke center (with MRI)	Yes	Routine pre-EVT MRI often available
Primary stroke center (transfer needed)	Often post-procedure	May not affect acute BP decisions
Mobile stroke unit	Limited	CT only typically

WMH Assessment Methods Beyond Fazekas: | Method | Measurement | Clinical Use | |———|————-|————| | **Fazekas scale** | Visual rating 0–6 | Research; clinical practice | | **Scheltens scale** | Volume-based | Research | | **WMH volume (automated)** | Quantitative mL | Research | | **ARWMC scale** | Regional ratings | Research |

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6. Supporting Evidence and Literature Context

6.1 Blood Pressure and Stroke Outcomes: Historical Context

BP Management in Acute Ischemic Stroke WITHOUT EVT:

Trial	Population	Comparison	Result
SCAST	n=2,029; acute ischemic stroke	Targets <140/90 vs 180/105	No benefit; possible harm
COSSACS	n=763; acute stroke	Continue vs stop prior antihypertensives	No difference
ENOS	n=4,011; acute stroke	Transdermal GTN vs control	No benefit
INTERACT-2	n=2,839; ICH	Intensive <140 vs <180 mmHg	Possible functional benefit in ICH

BP Management Post-EVT:

Study	Design	n	Finding
OPTIMAL-BP	RCT	307	Intensive BP worse outcomes (trial stopped early)
BP-TARGET	RCT	324	Intensive BP <130 mmHg increased white matter injury
ENCHANTED	RCT	2227	Intensive BP (target 130–140) vs standard (<140) — no difference in EVT subgroup
Observational studies	Multiple	Variable	J-curve relationship; both high and low BP associated with poor outcomes

6.2 WMH and Stroke Outcomes: Evidence Summary

WMH Predicts Worse Post-Stroke Outcomes:

Study	n	Finding
Arsava et al., 2009	295	Severe WMH associated with 3× higher probability of poor outcome
Chung et al., 2018	411	WMH volume independently predicts post-thrombolysis outcomes
Wang et al., 2020	623	Severe WMH associated with lower recanalization rates
Liu et al., 2023	1,247	WMH burden predicts hemorrhagic transformation post-EVT

6.3 Cerebral Autoregulation Monitoring

Emerging Role for Individualized BP Targets:

Technique	Parameter	Clinical Use
Transcranial Doppler (TCD)	CBF velocity reactivity	Research; limited clinical
Near-infrared spectroscopy (NIRS)	Cerebral oximetry	Operating room; emerging
ICP + MAP monitoring	CPP calculation	Neurocritical care standard

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7. Limitations and Future Directions

7.1 Study Limitations

Limitation	Impact	Mitigation
Secondary analysis	Not pre-specified as primary; hypothesis-generating	Pre-specified subgroup analysis; consistent with biological plausibility
Sample size	n=271; limited power for subgroups	Adequate for primary outcome; trends consistent
Open-label design	Potential bias in BP measurement	Blinded endpoint adjudication; objective primary outcome
Single ethnicity	South Korean population	Generalizability to other ethnicities unknown
MRI timing	WMH assessed pre-procedure; acute changes possible	Standard clinical practice; acute WMH changes unlikely in 24h
WMH only	Other CSVD markers not assessed	Fazekas is validated proxy for CSVD burden

7.2 Unanswered Questions

Question	Current Evidence	Needed Research
Optimal BP target in severe WMH	140–180 mmHg; harm <140 mmHg	RCT with WMH stratification
BP variability	High variability associated with worse outcomes	Optimal BP stability targets
Dynamic autoregulation	Static measures only	Continuous autoregulation monitoring
Long-term outcomes	90-day only	1-year functional outcomes
Other CSVD markers	WMH only	Microbleeds, lacunes, perivascular spaces
Generalizability	South Korea only	Multi-ethnic validation

7.3 Future Research Directions

1. **Prospective RCT with WMH stratification** — powered to detect difference in severe WMH subgroup
2. **Continuous autoregulation monitoring** — personalized BP targets based on real-time cerebral perfusion
3. **Multimodal CSVD assessment** — combining WMH, microbleeds, lacunes, perivascular spaces
4. **Imaging-based BP individualization** — CT or MRI perfusion to guide BP targets
5. **Long-term follow-up** — 1-5 year outcomes including cognitive function

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8. Conclusions and Expert Commentary

8.1 Summary of Evidence

The secondary analysis of the OPTIMAL-BP trial provides compelling evidence that pre-existing WMH severity significantly modifies the effect of intensive BP lowering post-EVT:

1. **Severe WMH (Fazekas ≥3) patients** receiving intensive BP control (<140 mmHg) had significantly worse functional outcomes (29.9% mRS 0–2) compared to conventional management (51.3%) — a 21.4 percentage point difference
2. **Mild WMH patients** showed no significant difference between intensive and conventional BP management (62.3% vs 68.5%), suggesting the harm is specific to the severely affected population
3. **Significant treatment effect modification** was confirmed (P = .008 for interaction), indicating the observed heterogeneity is not due to chance
4. **Biological plausibility** exists: impaired cerebral autoregulation in severe WMH renders the white matter vulnerable to hypoperfusion when BP is aggressively lowered
5. **Clinical implication:** Uniform intensive BP targets post-EVT are inappropriate; WMH severity should guide individualized BP management

8.2 Expert Perspective

> *"This secondary analysis challenges the simplistic 'lower is better' paradigm for post-thrombectomy BP management. The finding that severe WMH identifies a vulnerable population in whom aggressive BP lowering is harmful provides a mechanistic and clinical framework for individualized care."*

> *"Cerebral small vessel disease, as reflected by WMH burden, fundamentally alters the brain's capacity to autoregulate perfusion. In these patients, the 'one size fits all' approach to BP targets fails — and may cause harm."*

> *"This study underscores the importance of incorporating baseline brain health into acute stroke management decisions, much as we would consider age, comorbidities, or stroke severity."*

8.3 Take-Home Points for the Practicing Neurologist

1. **Avoid intensive BP lowering (<140 mmHg) in patients with severe WMH** post-EVT — this population showed 21% absolute increase in poor outcomes

2. **Target BP 140–180 mmHg in severe WMH patients** — consistent with current guidelines and the evidence from this analysis
3. **WMH severity modifies treatment effect** — severe WMH identifies patients who will be harmed by intensive BP control
4. **Mild WMH patients** may tolerate intensive BP control without significant harm, but routine intensive targeting is not supported
5. **"One size does not fit all"** — post-EVT BP management must be individualized based on baseline brain health
6. **Obtain baseline MRI when possible** — WMH assessment should be part of acute stroke workup when MRI is available
7. **Current guideline targets (140–180 mmHg)** appear appropriate for most patients, particularly those with severe WMH
8. **Ongoing uncertainty remains** — definitive RCT data with WMH stratification are needed to definitively establish optimal targets

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